

Module 24.02: The Dummy Variable Trap & Multicollinearity

1. The Phenomenon of Perfect Multicollinearity

When applying One-Hot Encoding to a categorical feature with K unique categories, generating K dummy columns creates a severe mathematical vulnerability known as the **Dummy Variable Trap**.

Consider a feature `Gender` with $K = 2$ levels: `Female` (D_1) and `Male` (D_2). For every observation i :

$$D_1^{(i)} + D_2^{(i)} = 1 \implies D_2^{(i)} = 1 - D_1^{(i)}$$

One dummy column is a deterministic linear combination of the other! This condition is known as **perfect multicollinearity**.

2. Mathematical Breakdown: Singularity of the Gram Matrix $\mathbf{X}^T \mathbf{X}$

Recall the closed-form Ordinary Least Squares (OLS) Normal Equation:

$$\mathbf{w}^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Where the feature design matrix $\mathbf{X} \in \mathbb{R}^{m \times (p+1)}$ includes an intercept column of ones:

$$\mathbf{X} = \begin{bmatrix} 1 & x_1^{(1)} & \dots & D_1^{(1)} & D_2^{(1)} & 1 & x_1^{(2)} & \dots & D_1^{(2)} & D_2^{(2)} & \vdots & \vdots & \ddots & \vdots & \vdots & 1 & x_1^{(m)} & \dots & D_1^{(m)} & D_2^{(m)} \end{bmatrix}$$

The Linear Dependence Proof:

Observe the relationship between the bias column $\mathbf{x}_0 = \mathbf{1}$ and the dummy columns:

$$\mathbf{1} - \mathbf{D}_1 - \mathbf{D}_2 = \mathbf{0}$$

Because the column vectors are linearly dependent:

1. The rank of the matrix is strictly deficient: $\text{rank}(\mathbf{X}) < p + 1$.
2. The Gram matrix $\mathbf{X}^T \mathbf{X}$ is **singular** (non-invertible).
3. The determinant is identically zero:

$$\det(\mathbf{X}^T \mathbf{X}) = 0$$

4. The inverse $(\mathbf{X}^T \mathbf{X})^{-1}$ does not exist!
 5. In numerical optimization libraries, this manifests as extreme numerical instability, near-infinite coefficient variance, or division-by-zero floating point exceptions.
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3. The Definitive Solution: The $K - 1$ Rule (`drop_first=True`)

To restore full rank to the design matrix, we must omit exactly **one** dummy indicator column for every categorical variable:

Number of Dummies Generated = $K - 1$

Original Feature: Region in ["Northeast", "Northwest", "Southeast", "Southwest"] ($K = 4$)

Omitted Category: "Northeast" (Baseline / Reference Category)

Retained Columns: D_nw, D_se, D_sw ($K - 1 = 3$ columns)

Observation is Northeast: D_nw = 0, D_se = 0, D_sw = 0 ---> Absorbed into Intercept b

Observation is Northwest: D_nw = 1, D_se = 0, D_sw = 0 ---> $b + w_{nw}$

Statistical Interpretation of the Intercept:

- When all retained dummy variables equal 0, the observation belongs to the omitted reference category.
- The intercept parameter b captures the expected baseline outcome of the reference group.
- The coefficient w_k represents the differential impact of category k relative to the omitted baseline group.